

Molecular Mechanism of Stabilizing the Helical Structure of Huntingtin N17 in a Micellar Environment

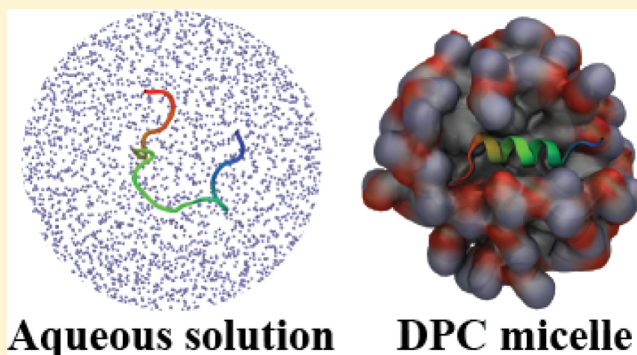
Leili Zhang,[†] Hongsuk Kang,[†] Francisco X. Vázquez,[†] Leticia Toledo-Sherman,[‡] Binqun Luan,[†] and Ruhong Zhou^{*,†}

[†]Computational Biology Center, IBM Thomas J. Watson Research Center, Yorktown Heights, New York 10598, United States

[‡]CHDI Management/CHDI Foundation, Los Angeles, California 90045, United States

S Supporting Information

ABSTRACT: Huntington's disease is a deadly neurodegenerative disease caused by the fibrilization of huntingtin (HTT) exon-1 protein mutants. Despite extensive efforts over the past decade, much remains unknown about the structures of (mutant) HTT exon-1 and their enigmatic roles in aggregation. Particularly, whether the first 17 residues in the N-terminal (HTT-N17) adopt a helical or a coiled structure remains unclear. Here, with the rigorous study of molecular dynamics simulations, we explored the most possible structures of HTT-N17 in both dodecylphosphocholine (DPC) micelles and aqueous solution, using three commonly applied force fields (OPLS-AA/L, CHARMM36, and AMBER99sb*-ILDNP) to examine the underlying molecular mechanisms and rule out potential artifacts. We show that local environments are essential for determining the secondary structure of HTT-N17. This is evidenced by the insertion of five hydrophobic residues of HTT-N17 into the DPC micelle, which promotes the formation of an amphipathic helix, whereas such amphipathic helices unfold quickly in aqueous solution. A relatively low free-energy barrier (~ 3 kcal/mol) for the secondary structure transformation was also observed for all three force fields from their respective folding-free-energy landscapes, which accounts for possible HTT-N17 conformational changes upon environmental shifts such as membrane binding and protein complex aggregation.



INTRODUCTION

Huntington's disease (HD) is a hereditary neurodegenerative disease caused by mutational expansion of CAG repeats in the exon-1 region of the huntingtin gene (Htt). The entire huntingtin (HTT) exon-1 protein (HTT exon-1) consists of three domains: an amphipathic region made of the first 17 residues (HTT-N17), a polyglutamine (polyQ) domain, and a polyproline-rich domain composed of 38 residues (HTT-C38). Structurally, HTT exon-1 and its mutants are intrinsically disordered proteins (IDPs) that can form amyloid-like fibers associated with HD once aggregated.¹

The biological function of the HTT-N17 domain has been explored in previous studies. Along with the polyQ domain in HTT exon-1, HTT-N17 is believed to affect the early development of human embryos.² Nuclear–cytoplasmic distribution of HTT exon-1 is strongly correlated to the toxicity of the HTT mutant, wherein nuclear exclusion is beneficial and nuclear inclusion is toxic.³ HTT-N17 was found to regulate nuclear localization of HTT exon-1 by post-translational phosphorylation of several residues in the domain.³ For example, in previous studies, the M8P mutation of HTT-N17 was found to block nuclear entry of HTT exon-1.⁴ As an amphipathic helix in a hydrophobic environment,⁵ HTT-N17 facilitates the targeting of the endoplasmic reticulum⁴ and mitochondrial membrane⁶ by

the protein. Knockout of HTT-N17 in zebrafish or mice induces HTT aggregation in the nuclei, resulting in a more rapid pathogenic phenotype.^{7,8} More recently, Crick et al.⁹ found that nonfibril aggregates of HTT exon-1 displayed the highest toxicity, wherein HTT-N17 destabilizes the nonfibril intermediates, resulting in more stabilized fibrils. On the other hand, another study shows that HTT-N17 significantly speeds up polyQ aggregation.¹⁰ However, how exactly HTT-N17 regulates the aforementioned nuclear inclusion, membrane targeting, and HTT exon-1 aggregation is still not well understood. Therefore, in light of the role of structure in the biological functions of Htt, the structural information of HTT-N17 is critical and invaluable in the field of HD research.

Previous experimental and computational studies have indicated that the structure of HTT-N17 in an amphipathic environment is mainly helical. For instance, Michalek et al.⁵ used NMR spectroscopy to solve the structure of HTT-N17 in dodecylphosphocholine (DPC) micelles (PDB-ID: 2LD2) as well as in 2,2,2-trifluoroethanol (TFE) solution (PDB-ID: 2LD0) and showed that it adopts a helical structure in

Received: February 15, 2017

Revised: April 13, 2017

Published: April 17, 2017

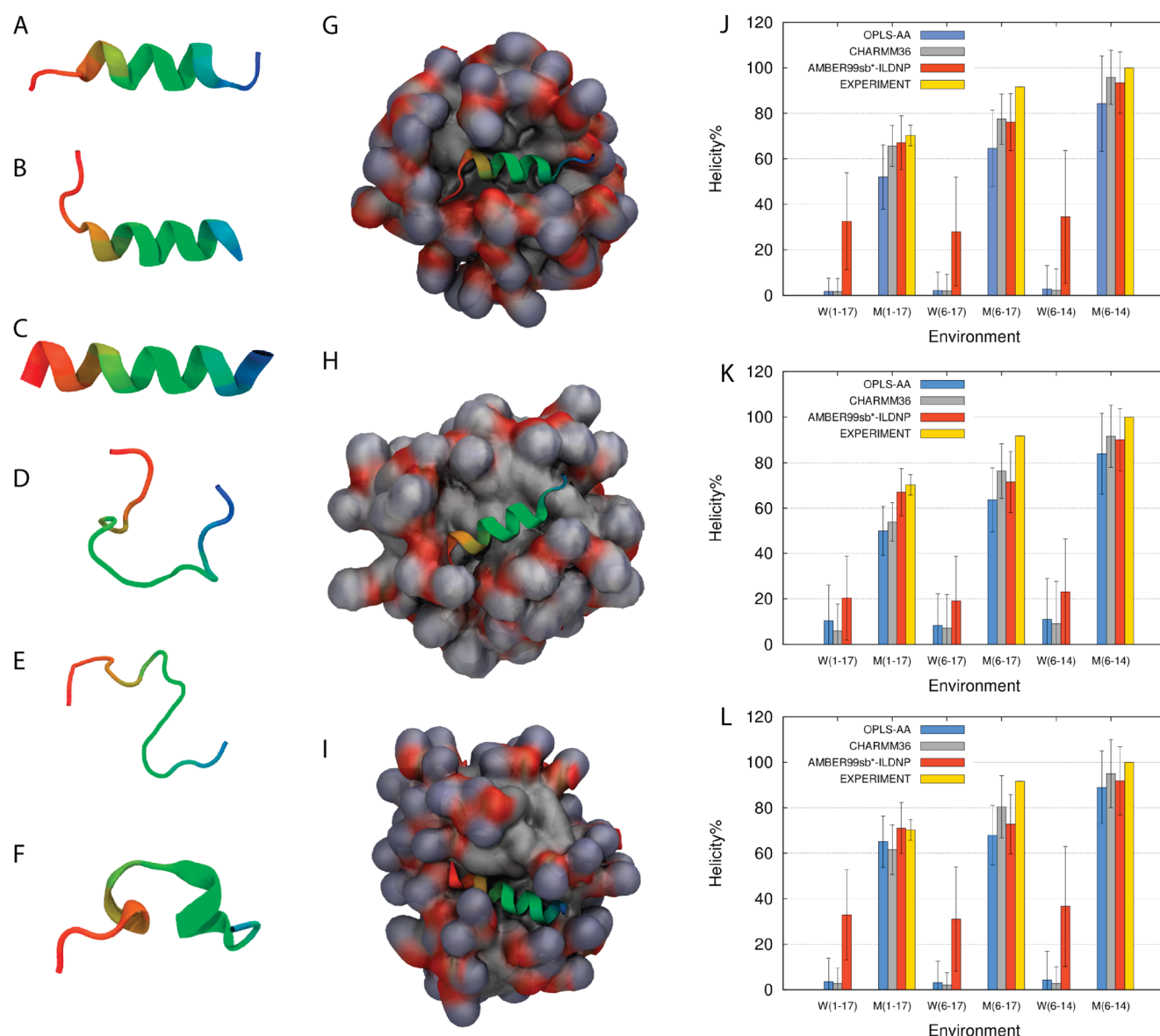


Figure 1. Representative structures and helicities of HTT-N17. Initial structures of (A) 2LD0, (B) 2LD2, and (C) **Helix**. Representative structures in water from different force fields: (D) OPLS-AA/L (OPLS-AA), (E) CHARMM36 (CHARMM), and (F) AMBER99sb*-ILDNP (AMBER). Representative structures on micelles from different force fields: (G) OPLS-AA, (H) CHARMM, and (I) AMBER, respectively. All proteins are colored with rainbow colors from the N-terminus (red) to the C-terminus (blue). DPC micelles are colored in gray (carbon) and red (oxygen). Helicities of HTT-N17 for different regions and different environments: 2LD0 (J), 2LD2 (K), and **Helix** (L). W stands for water and M stands for micelle. Numbers in parentheses stand for the residue ranges.

hydrophobic environments. It was also demonstrated that HTT-N17 is helical when co-crystallized with proteins such as maltose-binding protein (MBP)^{11,12} and single-chain Fv fragment.¹³ Also, Côté et al.¹⁴ performed simulations of HTT-N17 on a palmitoylphosphatidylcholine bilayer and achieved 85% helical content using the AMBER99sb*-ILDNP force field. However, stand-alone HTT-N17 in aqueous solution is speculated to be unstructured or in a random coil structure. For example, solution CD and NMR spectra strongly indicated unstructured HTT-N17 in water.¹⁰ Additionally, several computational research groups have studied the stability of helical HTT-N17 in an aqueous solution. Kelley et al.¹⁵ showed a 36% helical content of HTT-N17 starting from both the unstructured state and fully helical state in water solution using the AMBER03 force field. Rossetti et al.¹⁶ performed temperature replica exchange molecular dynamics (MD) simulations

and achieved 29% helical content of HTT-N17 in water using the AMBER99 force field. Binette et al.¹⁷ performed hamiltonian replica exchange MD simulations on HTT-N17 in water and obtained 29.3% helical content with the AMBER99sb*-ILDNP force field. Coincidentally, in all of the previous computational works on HTT-N17, some variations of the AMBER force field were selected for study (more below).

In this study, we conducted a systematic investigation of the molecular mechanisms of HTT-N17 under micellar and aqueous environments using all-atom MD simulations. To provide a benchmark and also avoid potential artifacts that arise from one particular force field, we explored three commonly used force fields, namely, OPLS-AA (OPLS-AA/L), AMBER (AMBER99sb*-ILDNP), and CHARMM (CHARMM36) for our simulations. AMBER99sb*-ILDNP was chosen over other AMBER variations because it has the least bias toward the

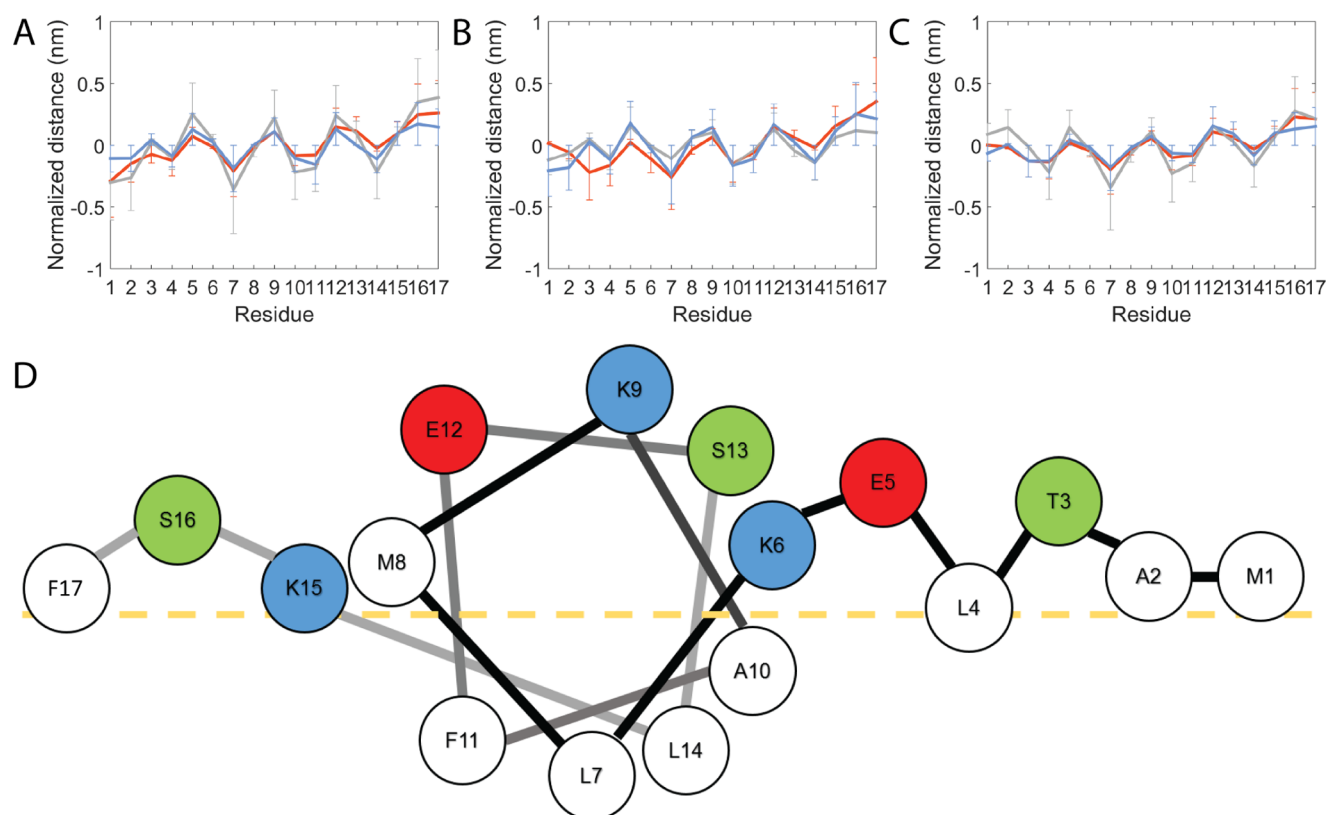


Figure 2. Relative insertion depth of the first 17 residues of HTT exon-1 (HTT-N17) residues in DPC micelles for three sets of simulations: (A) 2LD0; (B) 2LD2, and (C) **Helix**. Different colors represent the following: OPLS-AA (blue), CHARMM (grey), and AMBER (red). Data are collected from 150 to 400 ns in the corresponding simulations. (D) Illustration of the relative positions of each residue with regard to the membrane–water interface (orange dashed line). Hydrophobic residues are colored in white; noncharged hydrophilic residues are colored in green; positively charged residues are colored in blue; and negatively charged residues are colored in red. The helical part of HTT-N17 is prepared with pepwheel (<http://www.bioinformatics.nl/cgi-bin/emboss/pepwheel>).

helix. Also, we avoided using CHARMM27 because it has been known to cause problems in previous studies on IDPs. For example, Côté et al.¹⁸ showed that CHARMM27 results in significant overestimation of the HTT-N17 helical content in aqueous solutions (about 80%). It should be noted that Côté et al.¹⁸ also systematically explored the HTT-N17 peptide insertion mechanism and found that different placements of HTT-N17 can lead to different levels of irreversible destabilization of the helix, partly due to the difficulty of helix folding on membranes in the currently achievable time scales. In our current work, we focused on the “stabilization” of the helical content of HTT-N17 by the hydrophobic environment (using DPC micelles) with various widely used force fields.

Our results show that in the amphipathic environment all three force fields behave reasonably well by stabilizing HTT-N17 as an α -helix in a DPC micelle, which is in excellent agreement with the experimental data. As for the case of an aqueous environment, these force fields suggest the preference of a disordered coil structure for HTT-N17, albeit the fact that simulations using AMBER exhibit more helical content than that on using the other two force fields. Despite the force-field differences, we observe a universal small free-energy barrier of secondary structural change of HTT-N17. Our findings clarify the ambiguity in HTT-N17 structures reported in previous works and suggest a possible structural shifting mechanism of HTT-N17 in different environments.

METHODS

MD Simulation. We used GROMACS 5.0.5 for all simulations in this study. The OPLS-AA/L (OPLS-AA) force field¹⁹ was directly taken from the default GROMACS library. The AMBER99sb*-ILDNP (AMBER) force field²⁰ was obtained from the online library of the GROMACS website (<http://www.gromacs.org/>). The CHARMM36 (CHARMM) force field²¹ was obtained from the website of MacKerell labs (<http://mackerell.umaryland.edu/>). The united atom lipid force field by Berger et al.²² was adjusted according to the DPC molecule. We combined Berger’s lipid force field with OPLS-AA and AMBER based on previous studies.^{23,24} DPC micelles (containing 54 DPC molecules) were generated with the CHARMM-GUI website.²⁵ We utilized both virtual sites for the protein hydrogens and rigid H-bonds to enable a 4 fs time step for simulations with the OPLS-AA force field. Meanwhile, only the rigid H-bond option was enabled for simulations with the AMBER or CHARMM force field, to settle for a 2 fs time step. The N-terminus of the protein was capped by $-\text{NH}_3^+$, and the C-terminus was capped by $-\text{COO}^-$. Three initial structures of HTT-N17 (MATLEKLMKAFESLKSF) were prepared with different force-field setups: PDB-ID: 2LD0, PDB-ID: 2LD2,⁵ and **Helix** (artificially generated perfect α -helix). Following similar protocols to those used in our previous studies,^{26–29} all systems were solvated with TIP3P water and neutralized with a 100 mM NaCl solution. The initial structures for the micelle simulations are prepared so that HTT-N17 lies horizontally on the DPC micelle, with its hydrophobic residues facing toward the

micelle and its heavy atoms placed at least a few angstroms away from the DPC micelle. Production runs in the isobaric–isothermic ensemble (NPT) were performed following 50 000 steps of the steepest descent energy minimization and 10 ns of the constant volume, constant temperature (NVT) equilibration. Because of different secondary structure decay times (see the time-dependent plots for both helicity (Figure S1–S3) and radius of gyration (Figure S4–S6) provided in the Supporting Information), data were acquired over 400 ns from each of the micelle simulations and 1.2 μ s from each of the solution simulations. Structures were recorded with a 100 ps interval. Secondary structure analyses were performed using the STRIDE algorithm.³⁰

RESULTS

HTT-N17 Shows a Largely Helical Structure on DPC Micelles. As mentioned above, the helicity (defined as the percentage of residues being α -helices or 3–10 helix) of HTT-N17 has been disputed ever since the discovery of its importance to the HTT protein.^{5,10–13} Previous computational works have addressed this problem in various aspects,^{14–17} but none actually offer a direct comparison in different environments. Here, we started the simulations from the NMR structures of HTT-N17 (PDB-ID: 2LD0, PDB-ID: 2LD2) and an artificially generated perfect helix (referred to as 2LD0, 2LD2, and **Helix**, respectively, hereafter) in both aqueous solution and DPC micelles using three commonly applied force fields, OPLS-AA, AMBER, and CHARMM. The three initial structures used for the simulations are shown in Figure 1A–C, respectively. A quick examination of the HTT-N17 sequence shows its amphipathicity from the banded distribution of hydrophobic (**bold**) and hydrophilic (*italic*) residues: **MATLEKLMKAFESLKSF**. How these hydrophobic and hydrophilic residues contribute to its structural preference, particularly its helical content under different environments, is thus of great importance (and interest), given the complex nature of protein–protein interactions that HTT involves.

For the DPC micelles, HTT-N17 in all simulations converged toward similar helical structures on the micelle surface (Figure 1G–I). To show the orientation and insertion depth of each residue's side-chain in HTT-N17 on the micelle surface, we calculated the distance between the center of mass (COM) of each residue's side-chain and the COM of the entire micelle. To account for shape fluctuation of the micelle (e.g., deviation from a spherical one), that distance was further normalized by the distance between the COMs of HTT-N17 and the micelle, resulting in good agreement among all simulations regardless of the starting point and force field used (Figure 2A–C). Generally, negative values of normalized distances indicate that corresponding residue side-chains (e.g., hydrophobic ones) were inserted in the micelle, whereas positive values suggest that corresponding residues (e.g., hydrophilic ones) had their side-chains pointed away from the micelle surface.

An illustration of the relative positions of each residue with regard to the membrane–water interface is shown in Figure 2D. The end residues, M1, A2, and F17, for which secondary structures are not well defined and root-mean-square fluctuations (RMSFs) are significantly larger due to their end locations, are found to be more flexible than the rest of the peptide (Figure S7). Excluding these flexible terminal residues, we observed that the L4, L7, A10, F11, and L14 residues inserted into the hydrophobic center of DPC micelles, whereas the T3, E5, K9, E12, and S16 residues pointed toward water. The overall orientation thus

ensures maximal insertion of the hydrophobic residues into the hydrophobic micelle environment. In all, our simulated HTT-N17 structures agree well with the NMR structures⁵ and highlight the importance of the amphiphilic nature of the peptide. The details of the interactions involving HTT-N17 are further discussed in the section “Insertion Patterns of HTT-N17 in DPC Micelles”, with additional information on contact number analyses and insertion process analyses.

Additionally, we calculated the average helicity of HTT-N17 on the micelle surface, as shown in Figure 1J–L and Table 1. If

Table 1. Helicity of HTT-N17 in Different Systems with Different Force Fields^a

system	force field	helicity on micelle (%)	helicity in solution (%)
HTT-N17 (2LD0)	OPLS-AA	52.0 (14.1)	1.7 (5.9)
	CHARMM	65.7 (9.0)	1.6 (5.9)
	AMBER	67.1 (11.9)	32.6 (21.3)
HTT-N17 (2LD2)	OPLS-AA	50.0 (10.8)	10.5 (15.6)
	CHARMM	54.0 (8.5)	5.9 (11.8)
	AMBER	67.1 (10.4)	20.4 (18.5)
HTT-N17 (Helix)	OPLS-AA	65.1 (11.3)	3.5 (10.4)
	CHARMM	61.6 (10.8)	2.7 (6.9)
	AMBER	71.1 (11.2)	33.0 (19.9)

^aIn the system column, the starting structures in the simulations are listed in parentheses. In the helicity column, standard deviations are provided in parentheses.

starting from a perfect helix as the reference (Figures 1C and S9C), the OPLS-AA force field has a higher chance of forming a helix from residues 3 to 12. The CHARMM force field predicts a helix formed from residues 5 to 15, whereas AMBER predicts a helix formed from residues 3 to 12 (Figure S9C). The overall helicity is consistent among all simulations starting with all three initial structures (2LD0, 2LD2, and **Helix**), even though the detailed percentages can somewhat differ from one simulation to another. In short, all three force fields yielded high helicity values (50–70%), which agree well with the experimental values (calculated with crystal structure 2LD2). Therefore, our results of probing the helicity of HTT-N17 on micelles are comparable to the experimental data obtained under similar conditions (with the same lipid and same salt concentration).¹⁰

Helicity of HTT-N17 Is Significantly Reduced in Water.

In amphipathic environments, such as micelles and TFE solutions,⁵ or under crystallization conditions facilitated by fusing proteins, the HTT-N17 region adopts a predominantly helical structure.^{11–13} However, the HTT-N17 protein is also believed to be largely unstructured in aqueous solution.¹⁰

Representative structures of HTT-N17 in aqueous solution from all three starting structures (Figure 1A–C) are shown in Figure 1D–F. Here, both the OPLS-AA (D) and CHARMM (E) force fields yielded a random coil feature, resulting in <10% helicity, as seen in Figure 1J–L first column and Table 1. The AMBER (Figure 1F) force field also yielded a largely random coil but with a slightly helical feature, resulting in a higher helicity (20–30%). Not surprisingly, reducing the residue range for helicity calculations in the cases of OPLS-AA and CHARMM does not change the values (Figure 1J–L, third and fifth columns). However, the helicity slightly fluctuates in the case of AMBER when the range is changed to 6–17 or 6–14. This fluctuation comes from the bimodal distribution of helical residues shown in Figures 4B, S8B, and S9B.

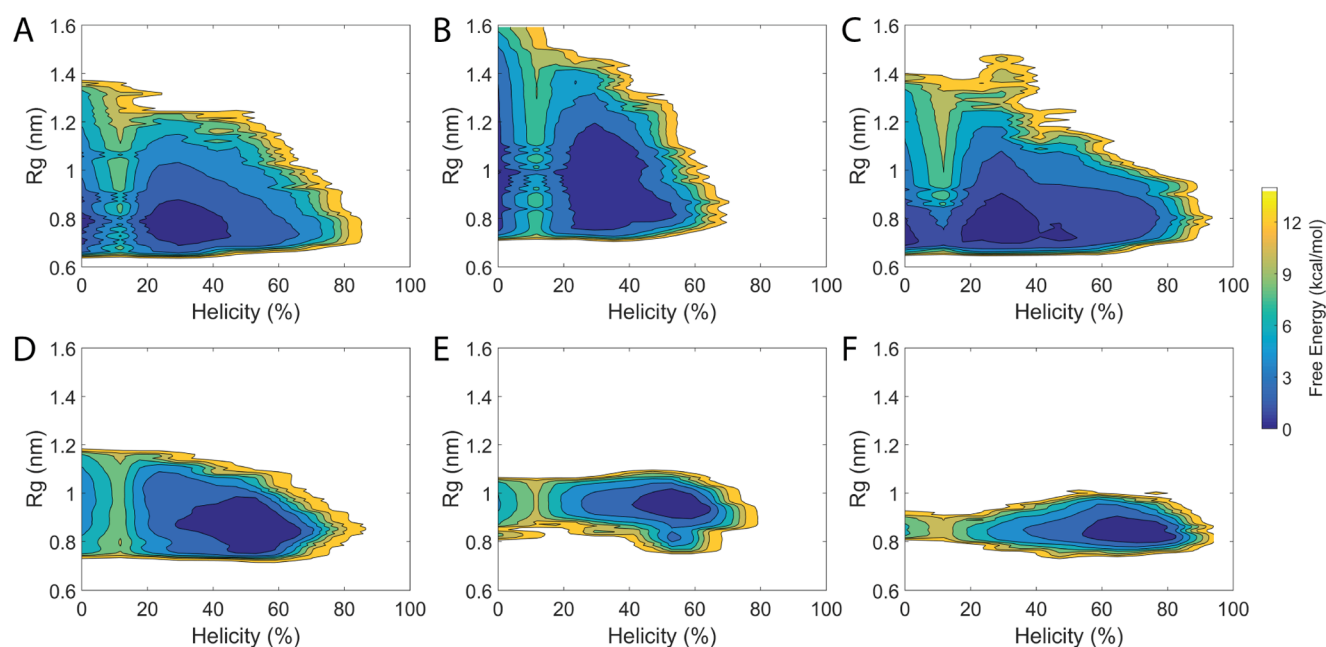


Figure 3. Relative free energy of HTT-N17 structures starting from 2LD2 with regard to the radius of gyration (R_g) and helicity. Different conditions are listed, as follows: (A) OPLS-AA, water; (B) CHARMM, water; (C) AMBER, water; (D) OPLS-AA, micelle; (E) CHARMM, micelle; (F) AMBER, micelle.

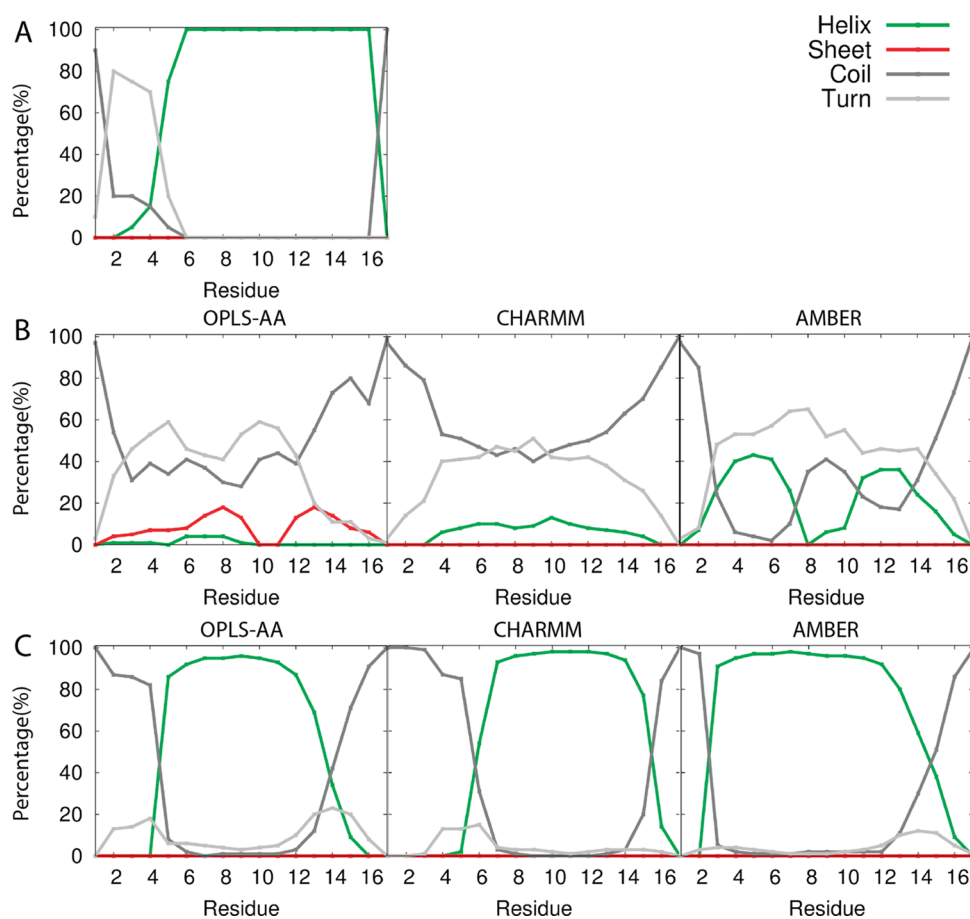


Figure 4. Relevant secondary structure features per residue for HTT-N17 under different conditions, starting from PDB-ID 2LD2. (A) 2LD2 crystal structure; (B) simulations in water; (C) simulations on micelles.

For comparison, we calculated the helicity difference (Δ_{helicity}) between HTT-N17 in aqueous solution and on micelles. We

obtained 50% for OPLS-AA, 57% for CHARMM, and 40% for AMBER (averaged over three different starting structures, with

data from Table 1). The significant helicity differences obtained with all three force fields unanimously indicate how the environment can bias HTT-N17 toward different secondary structures.

Free-Energy Landscapes Differentiate the Preference of Secondary Structures of HTT-N17 in Different Environments. To further stress that a subtle change in the surroundings may lead to significant structural changes of HTT-N17, we analyze the free-energy barrier of secondary structure transformations. Free-energy landscapes based on probability, shown in Figures 3, S10 and S11, can be written as

$$\Delta G = -k_B T \cdot \ln \frac{P_i}{\max(P_i)} \quad (1)$$

where state i denotes a certain range of radius of gyration R_g and helicity, ΔG is the relative free energy of the structure at the i th state, P_i is the probability of finding the system in the i th state, $\max(P_i)$ is the maximal probability, that is, the reference state, k_B is the Boltzmann constant, and T is the temperature. Similarity among the landscapes across three different starting points suggests a reasonable convergence of our simulations. The notable energy barriers present in all of the free-energy landscapes are due to the fact that at least two residues are needed to form a helix. From the free-energy landscapes of HTT-N17 on micelles, the lowest basins are always found where the helicity is roughly around 60%, as shown in Figure 3. Here, both CHARMM (Figure 3E) and AMBER (Figure 3F) force fields are seen to yield a more narrowed distribution along the R_g axis than OPLS-AA (Figure 3D). A wider distribution along R_g indicates a relatively higher flexibility of HTT-N17 on micelles with the OPLS-AA force field. When in water, the lowest basins of HTT-N17 using the OPLS-AA and CHARMM force fields are always on the R_g axis, which means that nonhelical structures are preferred. Meanwhile, for the AMBER force field, the lowest basins occur in the lower right part, suggesting that a wider distribution of helical structures is explored. This indicates that the collection of structures of AMBER is not limited to the singular structure provided by Figure 1F. Overall, the AMBER force field shows a greater preference toward helical structures compared with OPLS-AA and CHARMM, even though AMBER99sb*-ILDNP has the least bias toward helicity among all other AMBER variations.

As for the secondary structure transformation, Figure 3A–C shows that the structural spaces of HTT-N17 on micelles are generally well defined, with occasional “secondary-structure unwinding” where all dihedral angles deviate from the helical range defined by stride.³⁰ On the other hand, the structural spaces of HTT-N17 in water are much more widely distributed (Figure 3D–F), with mostly nonhelical structures. Interestingly, the free-energy barrier for secondary structure transition from a coil to a helix is less than 3 kcal/mol for all three force fields studied here. This indicates a rather flexible and versatile secondary structure of HTT-N17, which can be easily switched due to environmental changes, evidenced by the predominant helical basin in micellar environments. Consequently, we speculate that HTT-N17 adopts different structures upon fibril formation and cell membrane attachment.

Preferences of Different Secondary Structures Among the Three Tested Force Fields. β -Sheet formation plays an important role in HTT exon-1 aggregation. In our simulated HTT-N17, β -sheets were not observed for HTT-N17 on micelles (Figures 4C, S8C, and S9C). However, as seen in

Figures 4B, S8B, and S9B, the force fields show different preferences for secondary structure motifs in aqueous solution. The AMBER force field enhances helical structures in solution, especially for residues 3–7 and 11–14, displaying a bimodal distribution. The CHARMM force field remains largely unstructured in solution, with residual values of helicity. Surprisingly, there is a small amount of β -sheet content from residue 6–7 and 13–14 (Figure S9B) even when starting from a perfect helix. The OPLS-AA force field shows a small but consistent β -sheet content regardless of the starting point. The β -sheet content is seen to connect residues 2–4 and 14–16 or residues 7–9 and 12–14. The different preferences of secondary structures, due to different force fields, might affect the global protein structure in larger-scale simulations.

Insertion Patterns of HTT-N17 in DPC Micelles. We examined the contact numbers (with a cutoff value of 3 Å) between heavy atoms of HTT-N17 and DPC micelles in Figure S12. The insertion process is generally completed within the initial 50 ns of all simulations (partly due to the initial horizontal placements of HTT-N17 on DPC micelles wherein its hydrophobic residues face toward the micelles). The time scale for the peptide insertion is much smaller than the relaxation time of the protein’s secondary structure in water (~ 400 ns, see the time series of the helical content presented in Figures S1–S3). The relatively quick insertions here allow us to compare the stabilization of secondary structures of HTT-N17 in DPC micelles, starting from different initial structures.

There is a notable difference between the Berger force field (used by OPLS-AA and AMBER) and CHARMM force field, where the contact numbers are respectively stabilized at around 10 with the former compared to just around 5 with the latter. This indicates an overall deeper insertion of HTT-N17 into the Berger DPC membrane compared to that in the CHARMM membrane.

The detailed residue–lipid contacts and residue–micelle distances during the insertion process are shown in Figures S13 and S14, respectively. As clearly shown in Figure S13, residues T3, K6, and K15 dominate the initial contacts upon insertion. Distinctively, the Berger membranes prefer K9, whereas the CHARMM membranes prefer the M1 residue. Despite such differences, it is not surprising to see that for both membranes hydrophilic residues contribute significantly to the contacts because close contacts are mostly facilitated by hydrogen bonds and salt bridges.

In contrast to the dominating contributions to close contacts from hydrophilic residues, hydrophobic residues insert much deeper during this insertion process. Figure S14 shows the relative positions of residue side-chains during the insertion. Here, the relative positions of residue side-chains (with respect to the peptide COM) are defined as being the same as those in Figure 2 to avoid the effects from the fluctuations in micelle shapes. The first few residues that enter the DPC micelles are predominantly hydrophobic, including L7, A10, F11, and L14 in the Berger membranes and M1, A2, and L7 in the CHARMM membranes. These findings agree with the widely accepted amphipathic helix binding motif, where the hydrophilic residues provide electrostatic stabilization, whereas the hydrophobic ones bury deep into the membrane.

DISCUSSION

Since the discovery of the regulation functions of HTT-N17,^{4,6} there has been constant debate around its helicity in different environments from various studies. For example, experimentally,

HTT-N17 is found to be helical when crystalized in an amphipathic environment (micelles, TFE solution).⁵ Also, it is helical in a co-crystal in the MBP-bound state.^{11,12} However, the solution NMR spectra strongly indicate an unstructured conformation of HTT-N17 in water. From previous MD simulations, it was demonstrated that HTT-N17 remains stable as a helix in the lipid membrane environment,¹⁴ and it was predicted to be partially helical in solution with different variations of the AMBER force field.¹⁵ It is also notable that almost all secondary structure prediction servers, such as CFSSP (94.1%),³¹ GOR (41.2%),³² Jpred (94.1%),³³ NetSurfP (52.9%),³⁴ SOPMA (100%),³⁵ and SWISS-MODEL (100%),³⁶ generate very high scores on the helicity of HTT-N17. This might be due to the striped polar/hydrophobic alignment of residue types and preference of secondary structures of the residues. On the basis of these limited sources, one may argue that HTT-N17 is most likely helical in both environments: aqueous solution and lipid membranes.

Our findings reconcile the seemingly contradictory results and highlight that HTT-N17 indeed exhibits structural differences in aqueous and lipidic environments, suggesting the amphipathic helix formation to be an induced structure rather than being intrinsic. This is evidenced by the consistency observed across different initial structures and different force fields. Considering the multiple roles of HTT-N17, including nuclear entry and mitochondria binding,³³ regulations on HTT-N17, including SUMOylation and phosphorylation,^{37,38} and its enhancement of the overall filamentation of HTT exon-1,⁹ we believe that it adopts different structures in different states to mediate protein–protein interactions and protein–lipid interactions of HTT exon-1. This is supported by a rather low free-energy barrier (~ 3 kcal/mol) between unstructured HTT-N17 and helical HTT-N17, which can be biased or altered in different environments.

From a simulation point of view, force-field biases should be taken into consideration while evaluating the predicted structures of IDPs. Previous studies on IDPs have indicated some problems with force fields like CHARMM27 and AMBER03,³⁹ which we have deliberately avoided. Even then, we still discovered rather different biases on force fields in this investigation, unlike those in previous works.⁴⁰ Thus, precautions must be taken when examining the simulation studies carried out previously on HTT-N17. In detail, previous studies using AMBER force fields concluded that the helicity of HTT-N17 in aqueous solution is around 30%,^{15–17} which is reproducible in our simulations. However, given the largely unstructured NMR structure in aqueous solution, we believe this $\sim 30\%$ helicity number is significantly overestimated, similar to previous findings.⁴¹ On the other hand, both OPLS-AA and CHARMM (CHARMM36) force fields yield a reasonable helicity of $<10\%$ in aqueous solution. Nonetheless, the helicity differentiation (Δ_{helicity}) of HTT-N17 between the two environments simulated here are quite consistent among all three force fields: 50% for OPLS-AA, 57% for CHARMM, and 40% for AMBER, revealing that the secondary structure of HTT-N17 is indeed affected by the surrounding environments, with more helical content in a more hydrophobic environment. Our data with multiple force fields also provide a benchmark for further studies of this very important protein system.

Finally, despite the fact that the findings on HTT-N17 here reveal all of its possible structural space, the impact from trace amounts of β -sheet content is still unknown. Therefore, further investigations might be also helpful to evaluate the extent of such

transitions in different environments, similar to α -helix to β -sheet amyloid transitions in Alzheimer's disease.

CONCLUSIONS

In this study, we found a clear distinction in the secondary structures of HTT-N17 in different environments. Hydrophobic insertion of L4, L7, A10, F11, and L14 residues of HTT-N17 into the DPC micelle promotes the formation of an amphipathic helix, whereas such an amphipathic helix unfolds quickly in aqueous solution. These discoveries are in good agreement with the crystal structure of HTT-N17 on DPC micelles⁵ and the solution NMR data.¹⁰ A small free-energy barrier (~ 3 kcal/mol) between unstructured and helical HTT-N17 in water indicates the flexibility of this peptide, which may explain the structural ambiguity reported in the literature. The free-energy landscape of HTT-N17 changes significantly in DPC micelles. This may imply a folding upon the binding of HTT-N17 on membrane structures. Despite the fact that membrane structures of the cell are not needed for fibril formation,^{42,43} folding of HTT-N17 may indeed help localize HTT exon-1 in the cytoplasm and reduce the toxicity of HTT exon-1. Additionally, a trace amount of β -sheet content on HTT-N17 is found in aqueous solution. Such flexibility and secondary structural ambiguity could have a profound impact on the overall structure of HTT exon-1. Future studies focusing on the folding and unfolding of HTT-N17 in the full HTT exon-1 complex are highly desired.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpcb.7b01476.

Time-dependent plots of HTT-N17 helicity as well as time-dependent plots of the radius of gyration of HTT-N17, RMSF of HTT-N17 residues in DPC micelles, relative free energy of HTT-N17 structures starting from 2LD0 and **Helix**, secondary structural features per residue for HTT-N17 under different conditions starting from 2LD0 and **Helix**, contact numbers and contact numbers per residue (cutoff value: 3 Å) between HTT-N17 and the DPC micelle during the insertion process, relative positions of residues during the insertion process (PDF)

AUTHOR INFORMATION

Corresponding Author

*E-mail: ruhongz@us.ibm.com.

ORCID

Ruhong Zhou: 0000-0001-8624-5591

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

We would like to thank Joseph A. Morrone, Tien Huynh, and Michael Levitt for their help with this work. This work was partially funded by CHDI Foundation Inc., a charitable foundation that funds research on HD. R.Z. acknowledges the support of the IBM Blue Gene Science Program (W12S8591, W146412S, W1464164).

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